

National Environmental Data Management Plan

SECTION FOUR: PART I

Situational Analysis

Analysis of Current State of National Environmental Data



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List of Acronyms and Abbreviations

Acronym	Full Name
AQI	Air Quality Index
AUK	American University of Kuwait
BUR	Biennial Update Report
CAIT	Central Agency for Information Technology
CBD	Convention on Biological Diversity
CITRA	Communication and Information Technology Regulatory Authority
CO	Carbon Monoxide
CSB	Central Statistical Bureau
DGCA	Directorate General of Civil Aviation
EESD	Environmental and Earth Sciences Division (KISR)
EIA	Environmental Impact Assessment
EIS	Environmental Information System
eMISK	Environmental Monitoring Information System of Kuwait
EPA	Environment Public Authority (Kuwait)
FAO	Food and Agriculture Organization
FMIS	Financial Management Information System
GCC	Gulf Cooperation Council
GDP	Gross Domestic Product
GHG	Greenhouse Gas
GIS	Geographic Information System
GSSCPD	General Secretariat of the Supreme Council for Planning and Development
GUST	Gulf University for Science and Technology
HPC	High-Performance Computing
ICT	Information and Communication Technology

IEA	International Energy Agency
IUCN	International Union for Conservation of Nature
KGO	Kuwait Government Online (National Data Portal)
KIEIS	Kuwait Integrated Environmental Information System
KIN	Kuwait Information Network
KISR	Kuwait Institute for Scientific Research
KOC	Kuwait Oil Company
KPC	Kuwait Petroleum Corporation
KU	Kuwait University
MEA	Multilateral Environmental Agreement
MEW	Ministry of Electricity, Water and Renewable Energy
MSF	Multi-Stage Flash (desalination)
NAP	National Adaptation Plan
NDC	Nationally Determined Contribution
NEDMP	National Environmental Data Management Program / Plan
NKEC	National Knowledge Economy Center
NO₂	Nitrogen Dioxide
NO_x	Nitrogen Oxides
NUERS	National Unit for Environmental Research & Services (Kuwait University)
O₃	Ozone
PAAAFR	Public Authority for Agriculture Affairs and Fish Resources
PAAET	Public Authority for Applied Education and Training
PACI	Public Authority for Civil Information
PAHW	Public Authority for Housing Welfare
PAI	Public Authority for Industry
PM₁₀	Particulate Matter (10 micrometers or less)

PM_{2.5}	Particulate Matter (2.5 micrometers or less)
QA	Quality Assurance
QC	Quality Control
RO	Reverse Osmosis
ROPME	Regional Organization for the Protection of the Marine Environment
SAA-EPC	Shuaiba Area Authority - Environmental Protection Center
SDG	Sustainable Development Goal
SO₂	Sulfur Dioxide
SoE	State of Environment
UN	United Nations
UNDP	United Nations Development Programme
UNEP	United Nations Environment Programme
UNEP-ROWA	United Nations Environment Programme - Regional Office for West Asia
UNFCCC	United Nations Framework Convention on Climate Change
UNIDO	United Nations Industrial Development Organization
USGS	United States Geological Survey
VGI	Volunteered Geographic Information
WHO	World Health Organization
XBRL	eXtensible Business Reporting Language

1 Introduction

The Environmental Data Ecosystem in Kuwait has evolved over time to become a complex multi-institutional system where the various roles and responsibilities are spread across environmental protection authorities, research centers, ministries of economic sectors, as well as statistical centers. This development of the ecosystem has a long history going back several decades and spans the period of raising environmental awareness, rebuilding the country after the Gulf War, and establishment of the mandatory Environmental Protection Law No. 42 of 2014. Despite the fragmentation of roles and responsibilities in the ecosystem, which benefits from a multi-disciplinary expertise, it has been characterized by existence of some of gaps and weaknesses, such as redundancy, lack of inter-operability, among others. The National Environmental Data Management Program (NEDMP) aims at addressing some of these issues.

2 The Environmental Data Lifecycle

Data Lifecycle Model provides a robust framework for understanding how environmental data should be managed from collection through protection and sharing (USGS, 2013). The USGS model highlights that environmental data management is a cycle of six stages with three cross-cutting stages. The Kuwait National Environmental Data Management Plan (KNEDMP) has been structured to be aligned with the internationally recognised phases of the environmental data lifecycle to support the strategic objectives of the plan. The stakeholder's consultation meetings and assessment questionnaire provide the required data to analyse the existing environmental data lifecycle and sharing mechanisms.

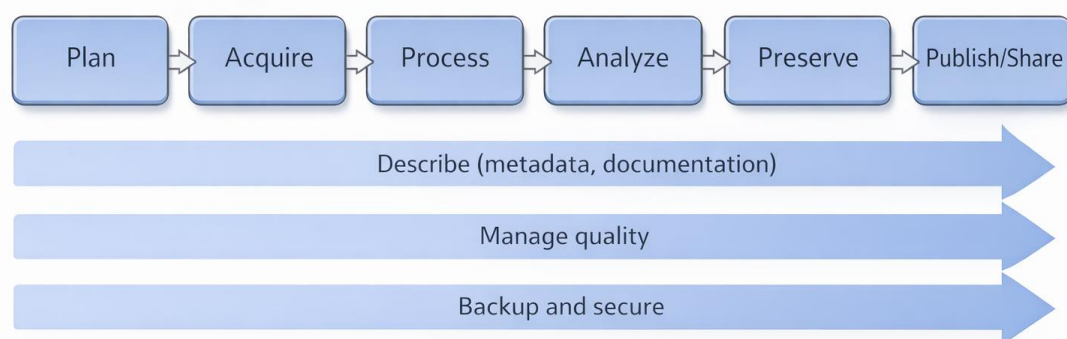


Figure 1 USGS Science Data Lifecycle Model

The NEDMP has adopted a lifecycle strategy for the management of the environmental data in Kuwait as a national resource. The strategy involves the collection, analysis, interpretation, presentation, archiving and preservation of the environmental data and information to be used for the formulation of environmental policies, for reporting to the United Nations Agencies and other international organizations and for meeting the requirements of the Kuwait Vision 2035.

2.1 Synergy Between NEDMP and Environmental data lifecycle stages

Plan: Data management plans will be developed to detail the roles of Kuwait national institutions for the NEDMP, current and proposed metadata standards, and any data sharing agreements that may need to be in place. This activity will set the stage for the realization of NEDMP Objective 1: Institutionalize Environmental Data Governance.

Acquire: Data collection using appropriate techniques and methodologies for the production of high-quality environmental data in support of NEDMP Objective 2: Ensure High-Quality and Standardized Environmental Data.

Process: Quality assurance validate and verify the field data collected under NEDMP Objective 2 and Pillar 2 (Data Quality).

Analyze: Data collection using appropriate techniques and methodologies for the production of high-quality environmental data in support of NEDMP Objective 2: Ensure High-Quality and Standardized Environmental Data.

Preserve: Data archiving through a robust infrastructure of reliable, sustainable long-term data repositories; aligning with NEDMP Objective 5: Build Sustainable National Capacities.

Publish/Share: Institutional interoperability to publish/share through the national data platforms/hubs to contribute to the achievement of the NEDMP Objective 4: Strengthen Data Sharing, Accessibility and Transparency and also to achieve SDG Target 16.10 (public access to information).

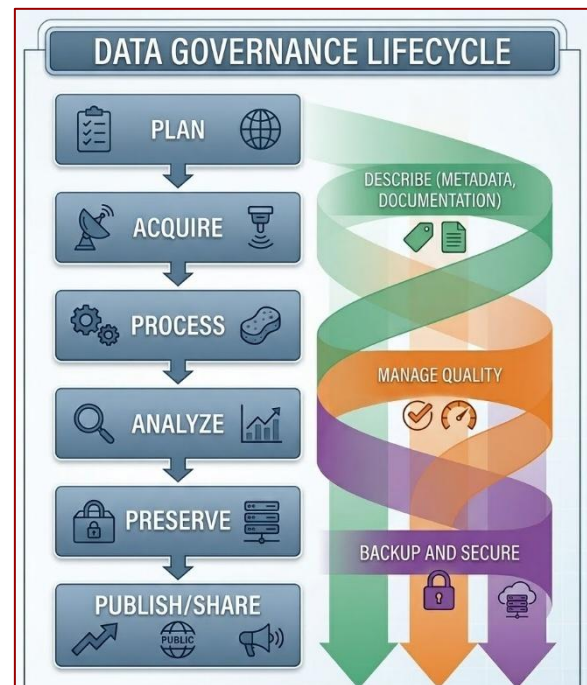


Figure 2 Data Governance Lifecycle

Table 1 alignment between the USGS Data Lifecycle stages and the NEDMP Strategic Objectives.

Model Stage	Relevant NEDMP Strategic Objective(s)
Plan	SO1: Institutionalize Governance
Acquire	SO2: Ensure Data Quality
Process	SO2: Ensure Data Quality SO3: Enable Interoperability
Analyze	SO4: Strengthen Data Sharing
Preserve	SO5: Build Sustainable Capacities

Publish/Share	SO4: Strengthen Data Sharing SO3: Enable Interoperability
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2.1.1 cross-cutting activities

To ensure accurate, reliable and protected data, three cross-cutting activities operate continuously across all data lifecycle stages:

Describe: Production of metadata to assist in data discovery and enhance the ability of data to communicate with other systems. Contributed to the achievement of NEDMP SO3: Enable Interoperability and System Integration.

Manage Quality: ensure that effective quality assurance and quality control mechanisms are in place for all activities carried out under the NEDMP, along the entire life cycle of the projects implemented under the programme, so that quality management is a process and is contributing to the achievement of SO2.

Backup and Security: this cross-cutting activity is directly linked with SO1 and SO5. It is also aligned with article 116 of Environmental Protection Law No. 42 of 2014 which states that data sovereignty is one of the principles and data must be safeguarded from loss, leakage, or unauthorized access (Government of Kuwait, 2014).

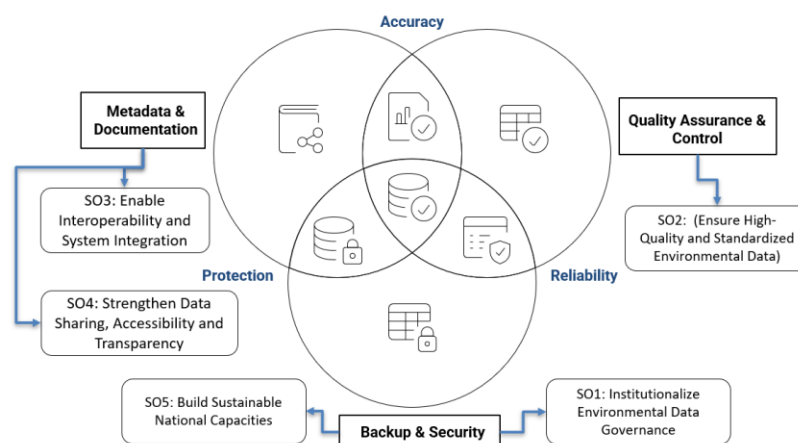


Figure 3 Alignment of Cross-Cutting Activities with NEDMP Strategic Objectives

2.2 Diversity of Environmental Data

The Environmental Data Stakeholders in Kuwait collectively produces vast amounts of environmental information, which are diverse in nature. Such information encompasses:

Thematic Environmental data: Air quality, physical, chemical and biological water quality parameters, marine sediment composition, thermal pollution measures. This category comprises physical environmental quality data including air quality parameters (e.g. particulate matter

concentrations), water quality indicators (e.g. temperature, pH, conductivity, dissolved oxygen levels, turbidity), marine sediment composition and thermal pollution measures.

Pollution source data: Data set for sources of pollution from industrial effluents, sewage and storm water discharges, oil spills and hazardous material transport.

Marine and coastal data: oceanographic characteristics (currents, waves, tides, salinity), bathymetry, sedimentation, and coastal zone morphology.

Biological and ecological data: Fisheries, Biological and ecological data set for the Fisheries research project: fisheries stock assessments, species distribution (fish, birds, coral reefs, plankton), nursery habitats and biological productivity.

Soil and landscape datasets: Soil classification, depth, analytical data, landform models, terrain analysis, landscape characterization, soil organic carbon estimates, agricultural soil and land suitability analysis for irrigation and agriculture, digital elevation models and hydrological modeling results, information on sediment yields and vulnerability to soil erosion, and data related to landslide risk and susceptibility.

Geospatial data: GIS layers covering main environmental domains, remote sensing imagery, and spatial maps.

Socioeconomic contextual data: including human demographics, distribution of labor, social and economic indicators, and health statistics.

3 Environmental Data Mapping

Environmental data mapping is crucial for the implementation of Kuwait NEDMP. The main objective is to understand the state of environmental data in Kuwait, specifically how it is distributed and located across different institutions. The Current State shows redundant data, lack of data interoperability, as well as missing information on crucial topics such as biodiversity, groundwater and climate change, which could be rectified if the environmental data such as air quality, marine monitoring, geospatial data, etc., were mapped systematically and effectively to eliminate redundancy, harmonize datasets, and enable interoperability for databases. Therefore, mapping environmental data enhances the institutionalization of environmental governance; ensuring that the environmental data is accurate and reliable, accessible, as well as sustainable, which are among the objectives of the NEDMP programme. Hence, data mapping for environmental assessment and for developing the state of environment report and other reporting obligations is key to realising these aims as they are aligned with Kuwait Vision 2035, as well as the requirements of reporting internationally on environment and sustainable development.

3.1 Environmental Monitoring Information System of Kuwait (eMISK)

The Environmental Monitoring Information System of Kuwait (eMISK) is the national environmental information platform developed by the Environment Public Authority (EPA) in

Kuwait for the purpose of environmental monitoring, data management and information-based decision making (Environment Public Authority, 2021).

The system has been established as an integrated environmental data infrastructure utilizing a range of GIS, database management application, analytical tools as well as web-based technologies. The eMISK hosts a substantial Environmental geodatabase having hundreds of spatial layers and data sets of various environmental significance covering Air, Water, Marine and Land use, Biodiversity, Energy, Waste & Industries etc.

3.1.1 eMISK Vision

“an internationally recognised centre of excellence for analysis and dissemination of environmental knowledge, providing authoritative strategic and timely information for the government, companies, researchers and the Kuwaiti public for development and implementation of policies and decisions for a sustainable environment” (Irfan-Ullah, 2012)

The data and tools integrated within the eMISK are used by the EPA and other relevant governmental organizations to provide quality information and tools to: Monitor environment performance, provide environmental indicators, facilitate policy formulation and decision making at all levels and ensure compliance with regulatory frameworks. (Environment Public Authority, 2021; Kuwait Government, 2023).

As an internal tool, eMISK is used as a data management system for the Environmental Public Authority (EPA) of Kuwait. Additionally, the system is also used for the provision of environmental information services to the public through the EPA portal and public applications that provide easy access to large amounts of environmental data, maps, reports and monitoring results.

These applications increase transparency about the environmental issues and promote public awareness and participation through an easy reporting mechanism for environmental complaints and pollution incidents.

With the integration of environmental monitoring systems, spatial data infrastructure and analytical dashboards, eMISK provides an important digital platform for environmental governance in Kuwait and contributes to the development of national environmental information systems and for informed environmental policy-making (Environment Public Authority, 2021; Fraunhofer UMSICHT, 2021)

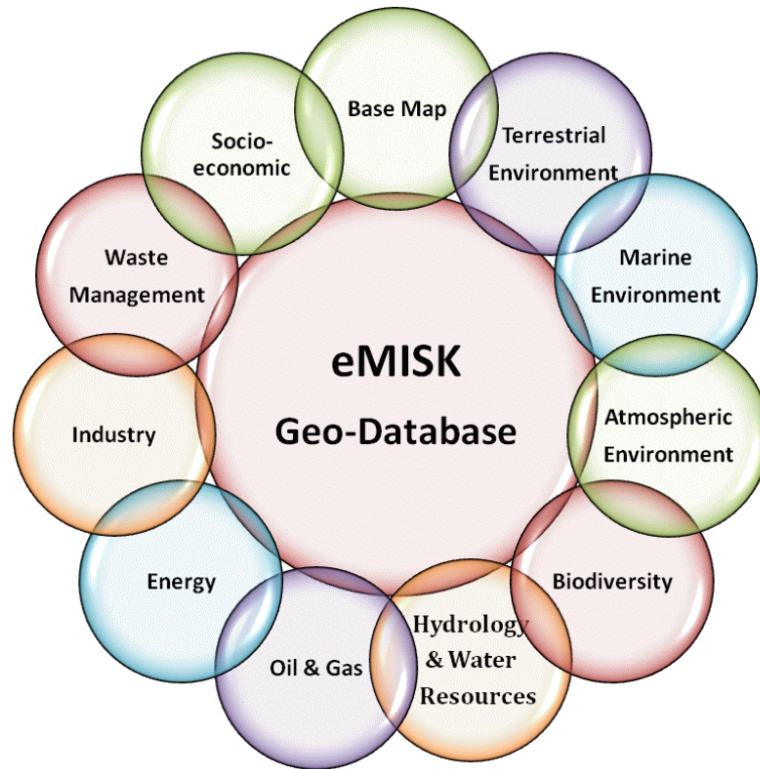


Figure 4 eMISK GeoDatabase domains (GISCON, 2018)

3.1.2 eMISK Objectives

Table 2 Objectives of eMisk

- Promote and disseminate environmental knowledge to all levels of the Kuwaiti society through a web portal
- Be a trusted national repository, source and a clearinghouse for geoenvironmental datasets
- Be a centre of excellence for training in geo-environmental analysis in the Region

Source: (Irfan-Ullah, 2012)

3.1.3 eMISK Data Sources

1. Government Entities: Kuwait EPA in addition to national authorities (providing GIS layers, air/water quality time series data).

2. Public Participation: Citizens providing data through the "Fix Our Environment" VGI tool (reporting problems and uploading geo-tagged photos).
3. Specialists/Experts: Subject matter experts who approve and validate incoming data.
4. Organizations: "several government, private and non-government organizations form an integral part of the information offered through Beatona."

Table 3 Summary of alignment between NEDMP and eMISK

NEDMP Objective	eMISK Objectives Aligned	Synergy Description
SO1: Institutionalize Governance	EM2 (Repository), EM4 (Database), EM7 (Decision Support)	eMISK already functions as a national repository under EPA leadership, embodying the governance structure NEDMP seeks to formalize
SO2: Quality & Standards	EM2 (Repository), EM4 (Database)	eMISK's geo-environmental database with 900+ parameters and 210 layers requires QA/QC frameworks that NEDMP's standards would formalize
SO3: Interoperability	EM2, EM4, EM5 (Beatona Portal)	eMISK's integrated database across 11 domains demonstrates interoperability that NEDMP aims to expand across ministries
SO4: Accessibility & Transparency	EM1 (Dissemination), EM5 (Beatona), EM6 (VGI), EM7 (Decision Support)	Beatona portal and "Fix Our Environment" VGI tool directly embody NEDMP's open data and public participation goals
SO5: Capacity Building	EM1 (Dissemination), EM3 (Training Centre), EM6 (VGI)	eMISK's vision as a "centre of excellence for training" and public engagement aligns with NEDMP's capacity building objective

NEDMP Objectives

eMISK Objectives

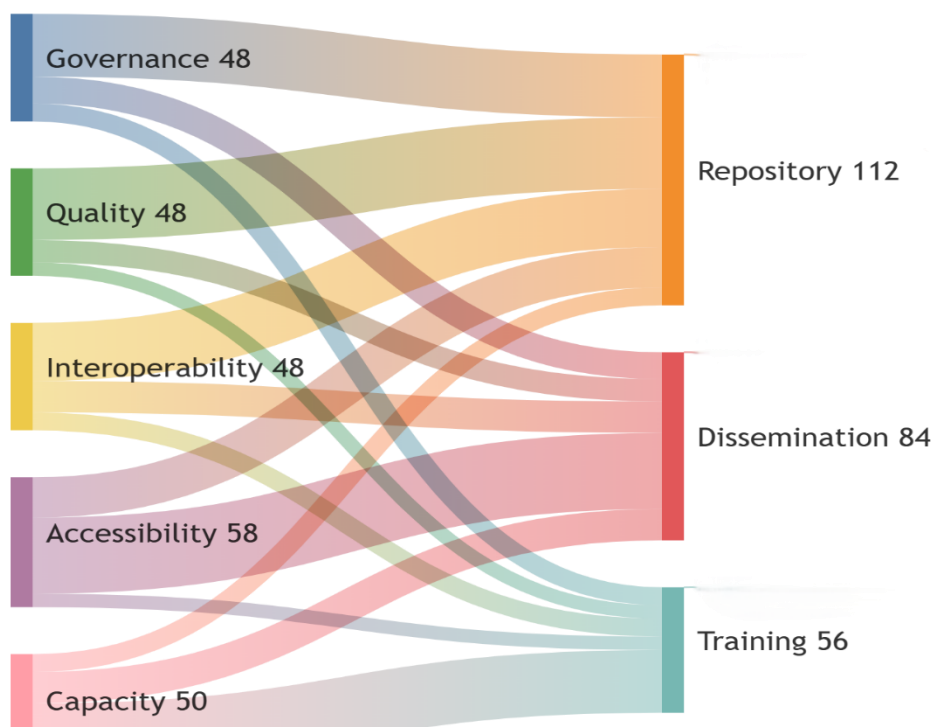


Figure 5 Synergy Mapping between NEDMP and eMISK

3.2 Beatona: Kuwait Official Environmental Portal

The beatona.net portal, is an environmental website for Kuwait managed by the Environmental Monitoring Information System of Kuwait (eMISK) of the Environment Public Authority (EPA).

The main purpose of the portal is to promote an environmental awareness to the public through sharing accurate scientific environmental information and data through an easy and simple online interface. The website also works as a front door to the eMISK geo-environmental database with additional data from ministries and other government organisations for ensuring accuracy and international standards.



Beatona portal presents real-time environmental statistics, educational information and a variety of resources including maps. The website enables public awareness and allows experts, as well as citizens to be informed about their natural environment. It assists policy makers to take informed decisions and be proactive towards the protection of the environment in the State of Kuwait.

3.2.1 Data Availability in Beatona Portal

Table 4 Availability of Data Within Beatona Web Portal

Data Category	Data Type	Key Parameters & Examples	Primary Source
Real-Time Monitoring	Air Quality (AQI)	PM2.5, PM10, NO2, SO2, CO, and O3 concentrations.	EPA Monitoring Stations
	Marine & Water	Seawater temperature, salinity, dissolved oxygen, and coastal bacterial counts.	KEPA Lab Analysis
	Meteorology	Ambient temperature, wind speed/direction, humidity, and pressure.	eMISK Integrated Sensors
Geospatial (GIS)	Interactive Maps	Land use, protected area boundaries (e.g., Jahra), and soil types.	eMISK Enterprise GIS
	Infrastructure	Locations of oil facilities, power plants, and industrial drainage outlets.	Multi-Agency GIS Layers
	Waste Management	Landfill locations, waste type distribution, and monitoring borehole data.	eMISK-Waste Module
Strategic & Scientific	National Reports	State of the Environment (SoE), National Adaptation Plans (NAP), and NDCs.	UNEP / EPA Publications
	Compliance Data	Stockholm Convention reports, Biennial Update Reports (BUR) for UNFCCC.	Environmental Data Dept.

	Technical Library	Scientific papers, environmental laws (Law 42/2014), and regulations.	EPA Legal Department
Public & Participatory	Crowdsourced (VGI)	Reports of oil slicks, dead fish sightings, or illegal dumping incidents.	Citizens / App Users
	Educational Media	Environmental magazines, awareness infographics, and species galleries.	EPA Awareness Dept.

3.2.2 Inventory of Requestable Environmental Data Types from Beatona

Table 5 Beatona Data Sharing Categories

Data Category	Description & Scope
Air Quality	Data from monitoring stations, including pollutant concentrations and air quality indices.
Marine Environment	Information on coastal water quality, marine habitats, and oceanic health indicators.
Terrestrial Environment	Data related to land use, desert ecosystems, and soil characteristics.
Biodiversity	Inventories of flora and fauna, protected area data, and wildlife sightings.
Water Resources	Statistics and quality assessments for groundwater and desalinated water systems.
Waste Management	Data on municipal, industrial, and hazardous waste volumes and disposal sites.
Industry	Environmental data related to industrial zones and compliance monitoring.
Oil and Gas	Environmental indicators specific to the petroleum sector and production areas.
Energy	Data concerning national energy consumption and its environmental footprint.
Base Maps	Geospatial GIS layers, topographic data, and foundational mapping for Kuwait.
Socio-Economic Data	Environmental indicators linked to population, public health, and economic activities.

4 Assessment of National Information Systems

Identifying national environmental information systems in Kuwait is crucial to enable data sharing and achieve interoperability as part of the NEDMP initiatives. The country's digital landscape is highlighted in the Kuwait Digital Landscape Whitepaper, a report by Microsoft, which describes the challenges and opportunities that Kuwait is currently experiencing. In relation to data management, a challenge lies in the fact that data is fragmented across government entities, and that this fragmentation is leading to poor decision-making due to lack of quality data. The report also points out that implementing "robust data governance frameworks that clearly define data governance models, data quality, as well as data protection and security" is essential to ensure that data is managed in a "consistent manner within the State of Kuwait (Microsoft, 2023, p. 14)". The Kuwait National Cybersecurity Strategy lists enhancing cooperation, coordination and information exchange between local and international entities within the cybersecurity sector as its third strategic objective (Government of Kuwait, 2017, p. 18). Furthermore, the Kuwait National AI Strategy Framework points out that "strengthening data and technology infrastructure to support AI will be necessary" through "comprehensive data governance policies and standards for the new era that takes into account all aspects including data security, privacy, quality, and accessibility (Microsoft, 2024, p. 17)". Identifying current national environmental information systems will help design a national environmental data portal that is highly integrated with the current systems through standardized interfaces to enable data exchange and AI capabilities. The overall aim is to have one single entry point for accessing environmental data while allowing current systems to operate with minimum disruption to the ongoing business processes. This will directly contribute to achieving NEDMP Strategic Objective 3: Enable Interoperability and System Integration, as well as Strategic Objective 4: Strengthen Data Sharing, Accessibility and Transparency.

4.1.1 Identification of Existing National Information Systems

Table 6 National Information Systems

System	System Topology	Responsible Entity	Data Types
Civil Registry System	Information System	Public Authority for Civil Information (PACI)	Demographic data, biometric data, vital records, addresses
National Environmental Information System (EIS)	Information System	Kuwait Institute for Scientific Research (KISR)	Marine water quality, bathymetry, oceanographic data, satellite imagery

Kuwait Information Network (KIN)	Infrastructure System	CAIT / CITRA	Network traffic data, inter-agency transactions, technical metadata
Kuwait e-Government Portal	Portal	CAIT	Transactional data, user profiles, government services data
eMISK (Environmental Monitoring Information System)	Information System	EPA	GIS layers, environmental monitoring data, air/water quality, biodiversity
Beatona Environmental Portal	Portal	EPA	Environmental indicators, AQI, reports, multimedia, awareness data
eMISK-Waste	Subsystem	EPA	Waste quantities, landfill data, transport logs, chemical data
K-Land (National Land Information System)	GIS System	Kuwait Municipality	Cadastral data, zoning, land use, infrastructure maps
Kuwait Integrated Environmental Information System (KIEIS)	Information System (Planned)	EPA	Unified datasets, environmental KPIs, compliance data
Client Inquiry Services System	Prototype System	Ministry of Commerce & Industry / UNDP	Contact center data, citizen service requests
Decision Support System	Prototype System	Council of Ministers	Aggregated inter-ministerial data
Developers Portal / Digital Integration Center	Portal	CAIT / Google Cloud	Inter-agency data exchange datasets

Unified Data Exchange Platform	Platform	PACI	Civil data, ID data, statistical data, verification data
Sahel Application	Mobile Platform	PACI / CAIT	Government service user data, integrated service data
IKARUS (HPC & AI Platform)	Advanced Platform	KISR	Simulation datasets, AI models, scientific and urban data
AI-Powered GIS / Digital Twin	GIS Platform	PACI	Geospatial data, digital twin datasets, infrastructure data
National Data Exchange Platform	Infrastructure Platform	CAIT	Interoperable government data exchange
Morasalat Online	Data Exchange System	CAIT / Ministry of Communications	Cross-government data exchange
National Data Portal (KGO)	Portal	CAIT	Open data, service data, government directory data
Civil Information System (PACI System)	Information System	PACI	Civil ID, demographics, residency data
Central Statistical System (CSB Databases)	Statistical System	CSB	National statistics, economic and socio-economic data
Financial Management Information System (FMIS)	Information System	Ministry of Finance	Budget, expenditure, revenue, financial data
Social Security Information System	Information System	Public Institution for Social Security	Pension, contributions, beneficiary data
Justice Information System	Information System	Ministry of Justice	Case data, court rulings, legal records

Security & Biometric System	Security System	Ministry of Interior	Criminal records, biometrics, residency/visa data
Kuwait Health Information System	Information System	Ministry of Health	Patient records, epidemiological data, health statistics
Education Information System	Information System	Ministry of Education	Student records, academic and administrative data
Traffic Information System	Information System	Ministry of Interior	Traffic violations, licensing, vehicle data
Immigration & Residency System	Information System	Ministry of Interior	Residency permits, visas, mobility data
Telecommunications & ICT Systems	Infrastructure System	CITRA	Telecom data, spectrum data, infrastructure data
Municipality GIS & Planning Systems	GIS System	Kuwait Municipality	Land use, planning, infrastructure spatial data
Public Authority for Housing Welfare Systems	Information System	PAHW	Housing applications, land allocation, infrastructure data
Customs Information System	Information System	General Administration of Customs	Trade data, import/export, declarations
Kuwait National Payment Gateway	Platform	Ministry of Finance / CAIT	e-payment transactions, financial data
National Cybersecurity & Data Infrastructure Systems	Infrastructure System	CITRA / National Cybersecurity Center	Cybersecurity, infrastructure monitoring data
Government Cloud Infrastructure	Infrastructure System	Ministry of Communications / CAIT / CITRA	Cloud-hosted systems, data storage services

National Bibliography Database (Kuwaitiana)	Database	National Library / NCCAL	Bibliographic data, publications, research records
Qaid Financial Filing System	Information System	Ministry of Commerce & Industry	Financial statements, structured XBRL data
Knowzone Scientific Platform	Platform	National Knowledge Economy Center	Scientific research databases, publications
Hawyti Digital Identity System	Identity System	PACI	Digital identity, authentication data
Labor Market Information System (LMIS)	Information System	CSB	Employment, labor force statistics
Air Quality Monitoring System	Monitoring System	EPA	Air pollution data, emissions, atmospheric data
Meteorological Data Systems	Information System	Directorate General of Civil Aviation	Weather, climate, atmospheric data
Oil & Energy Data Systems	Information System	KPC / Ministry of Electricity & Water	Energy production, emissions, water data
Agriculture & Fisheries Information Systems	Information System	PAAF	Agriculture, fisheries, biodiversity, land use data

5 National Environmental Data Stakeholders Mapping

As indicated earlier, the identification of national environmental data stakeholders is a crucial activity for the implementation of the NEDMP. This step is required to develop an understanding of the degree of fragmentation, as many environmental data stakeholders have been scattered among the environmental institutions, research centers, ministry departments and statistical bodies. The absence of a clear and detailed understanding of environmental data stakeholders and their respective roles and activities (i.e. data collection, management and use) could impact the NEDMP implementation and lead to dealing with ambiguities and complex problems. Therefore, mapping the key environmental data providers will shape the NEDMP intervention points to avoid data duplication, inconsistencies or lack of interoperability, and thus achievement of the desired objectives of the NEDMP will be guaranteed. Key stakeholders institutions are invited to participate in the NEDMP questionnaire and to get engaged in consultation meetings to

determine of the roles and responsibilities of the concerned institutions; data stakeholders' coordination and then reaching an agreement to share the data. This activity also aims at laying down the groundwork for building and achieving an interoperable national environmental information system that realizes its objectives and meets the requirements of Kuwait Vision 2035.

5.1 The Kuwait Environmental Public Authority (EPA)

The Kuwait Environmental Public Authority (EPA) is the main authority responsible for the protection of the environment. The EPA utilises the Environmental Monitoring Information System of Kuwait (eMISK) as well as the Beatona portal to manage a large geo-environmental database consisting of 11 sections, 39 sub-sections, 210 GIS layers and over 900 environmental variables. The data is categorised into air quality, water quality, marine water and sediment quality, industrial effluent, solid waste management and volunteered geographic information (VGI) provided by citizens, residents and schools through the "Fix Our Environment" portal. The EPA performs a dual role of being the operational authority responsible for monitoring and enforcement as well as the promotional body for dissemination of environmental knowledge and information to government agencies, research institutions, private sector companies and citizens.

5.2 Key Ministries and Governmental Agencies

In addition to the Environment Public Authority (EPA), sectoral ministries and other national authorities generate environment-related data as a byproduct of their core mandate. Data on oil activities including exploration, spills, refinery and petrochemical operations, as well as natural oil seeps have been generated by the Ministry of Oil, because of the petroleum-based economy of Kuwait.

The Ministry of Electricity, Water and Renewable Energy (MEW) also generates data on power and desalination operations, including cooling water releases and thermal pollution, which are related to water and electricity production. In addition, the Public Authority for Agriculture Affairs and Fish Resources (PAAAFR) has generated fisheries-related data on fish stocks assessment, aquaculture, and the use of marine resources. The Ministry of Health generates environment-related health data, analysis of morbidity and mortality rates that can be used in conjunction with environmental indicator data.

Additionally, there are governmental sources that are providing valuable environmental-related data including Kuwait Municipality for geodata which provides urban environmental indicators, waste management. Governmental data providers include Shuaiba Area Authority - Environmental Protection Center (SAA-EPC) for industrial zone monitoring data, Ministry of Public Works for infrastructure environmental data and Kuwait Ports Authority for marine transportation, dredging, hazardous materials data. Directorate General of Civil Aviation (DGCA) provides data for meteorology and aviation emissions.

5.3 National Knowledge Economy Center (NKEC)

National Knowledge Economy Center (NKEC) is a unit in the General Secretariat of the Supreme Council for Planning and Development (GSSCPD) affiliated institutions in Kuwait. NKEC is one of the key strategic enablers for achieving Kuwait Vision 2035. NKEC formulates national strategic plans, and influences government policies to enhance the process of innovation, entrepreneurship, and digitalization. In line with its objectives, the center is charged with the responsibility of developing and executing a number of national programs. Notably, it developed the Scientific Search Engine known as (Knowzone) which aims to grant researchers and the public access to international scientific research databases and specialized scientific journals to support research activities and better facilitate the process of taking informed and scientifically-backed decisions.

Table 7 Data use Beatona - National Knowledge Economy Center

Goal	Beatona (EPA)	NKEC (GSSCPD)
Data Type	Raw Data (AQI, Water Quality, Maps).	Research Data (Academic papers, Policy briefs).
Focus	Current Environmental Status.	Future Strategic Direction & Knowledge.
Output	Real-time monitoring & violation reports.	National development plans & scientific literature.

5.4 Key Academic Institutions

The Kuwait Institute for Scientific Research (KISR) is a key academic source of environmental scientific research and data. Since 2002, KISR has been enhancing its capabilities in developing environmental information systems (EIS) and in 2003 it started with EIS project dealing with the marine environment and industrial effluents. KISR's data are provided by the following divisions: - Environmental and Earth Sciences Division (EESD) deals with collecting oceanographic data, geological data, fisheries and oil pollution data, as well as sedimentation data, in addition to species distribution data and remote sensing. In contrast to the EPA's regulatory mission, KISR carries out a research mission that provides essential scientific data to support national policy-making and address existing scientific gaps in the State of Kuwait.

Kuwait University (KU): Considered one of the main sources for environmental data in Kuwait with a large number of laboratories and centres such as:

- National Unit for Environmental Research & Services (NUERS): it provides laboratories services and facilities for data analysis, covering fields such as chemical and environmental analysis to researchers, institutions, organizations and businesses.
- Department of Earth and Environmental Sciences: it is engaged in scientific research and studies on local geological and environmental trends.

- The College of Life Sciences: is providing undergraduate programs of study such as the BSc Environmental Science, with a focus on environmental information systems and resource conservation.

The Public Authority for Applied Education and Training (PAAET): has specialized vocational and technical programs, such as the College of Technological Studies, which is a source of data capacity building programmes through applied research in engineering and environmental technology.

The Private Academic Institutions: The American University of Kuwait (AUK) and the Gulf University for Science and Technology (GUST) have organized international conferences and research projects relating to sustainability and environmental diplomacy in the Gulf region.

5.5 Key National Organizations

National Organizations are considered additional contributors: - Kuwait Municipality for geodata, urban environmental indicators and waste management, Shuaiba Area Authority-Environmental Protection Center (SAA-EPC) for industrial zone monitoring, Ministry of Public Works for infrastructure-related environmental data, Kuwait Ports Authority for marine transportation, dredging and hazardous materials, Directorate General of Civil Aviation (DGCA) for meteorological and aviation emissions data.

5.6 Key Data Aggregators and Custodians

There are a number of national data aggregators and custodians including the Central Statistical Bureau (CSB) which provides source data on population, demographics, health, labour and other economic and education data which are a pre-requisite for environmental analysis and Integrated Environmental Assessment (EIA). The Central Agency for Information Technology (CAIT) provides the policies, standards and regulations for geodata as well as ICT infrastructure and data sharing and interoperability frameworks. The Communication and Information Technology Regulatory Authority (CITRA) regulates telecommunications and internet infrastructure to facilitate online environmental knowledge sharing.

5.7 Environmental Data Partners in Kuwait

Table 8 Key Environmental Data Stakeholders

Institution	Primary Role	Key Environmental Data Types
Environment Public Authority (EPA)	Environmental Regulator	Air quality (PM2.5, PM10, NO2, SO2, CO, O3), water quality, marine water and sediment quality, industrial effluents, solid waste management, GIS layers (210 layers), environmental parameters (900+ parameters), volunteered geographic information (VGI), State of Environment (SoE) reports, National Adaptation Plans (NAP), NDCs, compliance data (Stockholm Convention, UNFCCC BURs), environmental laws and regulations

Kuwait Institute for Scientific Research (KISR)	Scientific Research	Marine environment data, oceanographic characteristics, geological data, fisheries data, oil pollution data, sedimentation data, species distribution data (fish, birds, coral reefs, plankton), remote sensing data, environmental information systems (EIS) research, industrial effluent data
Ministry of Oil	Oil & Gas Regulator	Oil exploration data, oil spills data, refinery operations data, petrochemical production data, natural oil seeps data, emissions data
Ministry of Electricity, Water and Renewable Energy (MEW)	Water & Energy	Power generation data, desalination plant data, cooling water discharge data, thermal pollution data, water resources data
Ministry of Health	Health Regulator	Environment-related health data, mortality data, morbidity data, health statistics correlated with environmental indicators
Public Authority for Agriculture Affairs and Fish Resources (PAAAFR)	Agriculture & Fisheries	Fisheries stock assessments, aquaculture data, marine resource utilization data, agricultural data
Kuwait Municipality	Municipal Authority	Geodata, GIS data, urban environmental indicators, waste management data, land use data
Shuaiba Area Authority - Environmental Protection Center (SAA-EPC)	Industrial Zone Monitoring	Industrial effluents data, air quality data (industrial zone), marine discharge data, industrial zone environmental monitoring data
Ministry of Public Works	Infrastructure	Infrastructure environmental data, environmental protection center data, engineering environmental data
Kuwait Ports Authority	Maritime Authority	Marine transportation data, dredging data, hazardous materials transport data, port operations data
Directorate General of Civil Aviation (DGCA)	Aviation Authority	Meteorological data (ambient temperature, wind speed/direction, humidity, pressure), aviation emissions data, air transportation data
Central Statistical Bureau (CSB)	Data Aggregator / Custodian	Population data, demographic data, health statistics, labor force data, economic data, education data
Central Agency for Information Technology (CAIT)	IT Governance	Geodata standards, ICT infrastructure data, data governance frameworks, data sharing and interoperability frameworks, policies and regulations for geodata

Communication and Information Technology Regulatory Authority (CITRA)	Telecom Regulator	ICT infrastructure data, internet access data, telecommunications infrastructure data
National Knowledge Economy Center (NKEC)	Strategic Enabler / Knowledge Hub	Scientific research databases, academic papers, policy briefs, scientific journals access (Knowzone platform), knowledge management for Kuwait Vision 2035
Kuwait University (KU)	Academic Institution	Environmental research data, chemical and environmental analysis data (via NUERS), geological and environmental studies data, environmental information systems research
National Unit for Environmental Research & Services (NUERS)	Research Services	Laboratory services data, chemical analysis data, environmental analysis data for researchers and institutions
Department of Earth and Environmental Sciences (KU)	Academic Research	Geological data, environmental trends data, scientific research studies
College of Life Sciences (KU)	Academic Education	Environmental science education data, environmental information systems research, resource conservation data
Public Authority for Applied Education and Training (PAAET)	Vocational & Technical Education	Applied research data in engineering, environmental technology data, capacity building programs
College of Technological Studies (PAAET)	Technical Education	Applied research data, engineering environmental data, technical capacity building
American University of Kuwait (AUK)	Private Academic Institution	Sustainability research data, environmental diplomacy research, international conference outputs
Gulf University for Science and Technology (GUST)	Private Academic Institution	Sustainability research data, environmental diplomacy research, regional environmental studies
General Secretariat of the Supreme Council for Planning and Development (GSSCPD)	National Planning Authority	National strategic plans, Kuwait Vision 2035 implementation, development policy frameworks
Ministry of Interior	Security & Enforcement	Land transportation data, coastal zone enforcement data, traffic-related environmental impacts (inferred from other documents)

6 Data Required for Integrated Environmental Assessment

Environmental Data is one of the key elements that contributes to Kuwait Integrated Environmental Assessment. The State of the Environment Report 2024 is an example of how high quality multi-sourced data that can be used as the basis for the environmental assessment and analysis of the driving forces, pressures, the state of environment, and the impact of policies towards assessing the sustainability of environment in the country. Comprehensive and reliable timely data, is necessary for achievement of Kuwait Vision 2035 goals and for implementation of the country's international environmental commitments. In this regard, the data has been more than an evidence base for the environmental assessment; it is considered a backbone tool for decision-making for sustainable management of the environment and future planning for development.

6.1 Kuwait State of The Environment Report 2024

The State of Environment Report 2024 is Kuwait's first State of the Environment (SoE) report. The Environment Public Authority (EPA) developed this comprehensive evidence-based report in collaboration with the United Nations Environment Programme (UNEP). Based on a data-driven approach and grounded in the integrated environmental assessment framework of UNEP, this SoE report assesses in a systematic manner the current state of the environment in relation to the seven environmental themes namely: Climate Change, Solid Waste, Land, Water, Atmosphere, Biodiversity and Coastal and Marine Environment. The report emphasizes the importance of data as the basis and resource for environmental protection. It identifies data gaps, where for instance, data on air emissions of certain hazardous air pollutants is lacking, marine monitoring stations are inadequate, and a national solid waste database is not yet established. Rather than just recording these gaps, the report addresses them as a necessity that has to further enhance the performance of environmental governance in Kuwait. In addition, the report emphasizes the necessity of having adequate, quality, accurate, reliable, and integrated data in order to meet the country's obligations in relation to international environmental agreements, achieve the targets of Kuwait Vision 2035 and SDGs, as well as to facilitate the development of policies based on scientific and data-driven evidence.

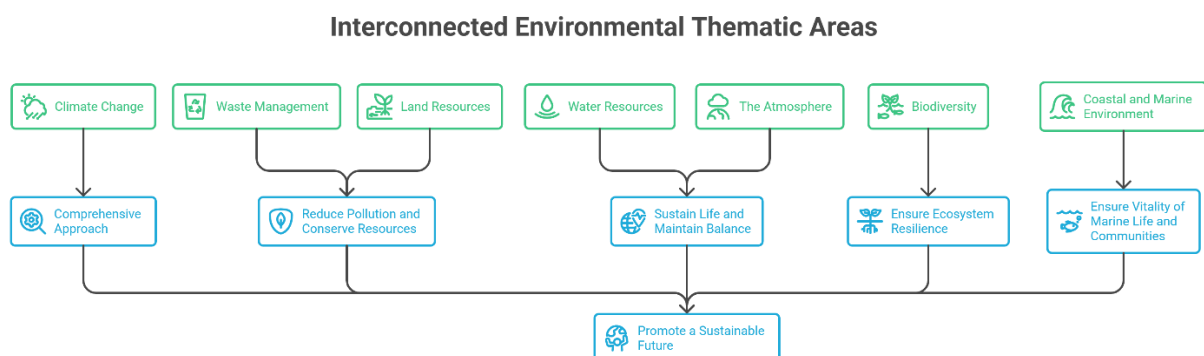


Figure 6 Thematic Areas of the SOE Report

6.1.1 Types of Data Required for the State of Environment Report

Table 9 Data Required for the State of Environment Report

Theme	Data Type	Description / Purpose	Source Type
1. Drivers of Environmental Change	Economic data	GDP by sector, oil and gas revenues, exports, government revenues, per capita income	National (CSB, IEA)
	Demographic data	Population size, population growth rates, citizenship breakdown (Kuwaiti vs. expatriate), age distribution, urbanization rates	National (CSB)
	Consumption data	Electricity consumption by sector, water consumption by sector, private vehicle ownership trends	National (MEW, Ministry of Interior)
	Climate and geographical data	Average annual precipitation, evaporation rates, temperature trends, wind patterns	National (DGCA, Meteorological Department)
2. Climate Change	Greenhouse gas emissions data	CO ₂ and other GHG emissions by sector (energy, electricity, water desalination, industry), historical emissions trends	National (EPA), International (UNFCCC reports)
	Temperature projections	Historical and projected temperature rise (e.g., up to 4.5°C above historical levels)	National (second national communication to UNFCCC)
	Sea level rise data	Sea level measurements, projections for coastal inundation, impact assessments for islands (e.g., Bubiyan)	National (KISR), EPA monitoring stations
	Energy sector data	Installed power generation capacity, fuel mix for electricity and desalination, flare gas recovery data	National (MEW, Kuwait Oil Company)
	Adaptation initiative data	Renewable energy projects (e.g., Shagaya plant), coastal protection measures, marine protected areas	National (EPA, KISR, Ministry of Health)
3. Waste Management	Solid waste data	Total annual solid waste generation (tons), municipal waste per capita, waste composition (organic, recyclables, etc.)	National (CSB, EPA, Kuwait Municipality)

	Landfill data	Number of landfills (active and closed), area covered, liner systems, leachate and gas collection systems status	National (EPA, Kuwait Municipality)
	Liquid waste data	Sewage and industrial liquid waste volumes, treatment capacity, reuse rates, discharge volumes into sea	National (Ministry of Public Works, Public Authority for Industry)
	Hazardous waste data	Medical waste, industrial hazardous waste, used oils, batteries, chemicals, electronic waste volumes	National (Ministry of Health, EPA, licensed companies)
	Recycling data	Recycling rates, number of recycling plants, types of materials recycled, products from recycled materials	National (EPA, Public Authority for Industry)
	International convention data	Compliance data for Minamata, Basel, Stockholm Conventions	International (UN conventions)
4. Land Resources	Land use/land cover data	Urban expansion, housing, roads, infrastructure land use (2000-2016), camping areas, nature reserves	National (EPA, Kuwait Municipality, CSB)
	Sand and dust storm data	Frequency, intensity, spatial extent, sources (domestic and transboundary), composition of dust particles	National (Meteorological Department, EPA, KISR)
	Soil degradation data	Wind erosion rates, soil compaction from vehicle traffic, soil salinity, fertility loss	National (KISR, EPA, academic studies)
	Agricultural data	Chemical and organic fertilizer use, irrigation practices, agricultural waste management	National (CSB, PAAF)
	Desertification data	Implementation status of UNCCD, national action plans, restoration projects	National (EPA), International (UNCCD)
5. Water Resources	Freshwater resources data	Renewable water availability per capita, groundwater levels, aquifer salinity, extraction rates	National (MEW, KISR)
	Desalination data	Freshwater production volumes (million m ³ /year), number and type of desalination plants (MSF, RO), energy consumption for desalination	National (MEW)

	Water consumption data	Per capita consumption (liters/person/day), sectoral consumption (domestic, agriculture, industrial)	National (CSB, MEW)
	Wastewater data	Treated wastewater volumes, reuse rates, irrigation distribution networks, discharge into sea	National (Ministry of Public Works)
	Water pricing and subsidies data	Water tariff structure, subsidy levels, cost recovery rates	National (MEW, Ministry of Finance)
6. Atmosphere	Air quality monitoring data	Concentrations of PM ₁₀ , PM _{2.5} , SO ₂ , NO _x , O ₃ , CO; Air Quality Index (AQI) from monitoring stations	National (EPA fixed monitoring network, Kuwait Oil Company stations)
	Emissions inventory data	Emissions from power plants, oil refineries, petrochemical industries, transportation, industrial activities	National (EPA, KISR, Harvard School of Public Health collaboration)
	Vehicle data	Number of licensed vehicles, vehicle types, fuel consumption, emissions from transportation sector	National (Ministry of Interior, General Department for Information Systems)
	Meteorological data	Wind speed and direction, temperature, humidity, atmospheric stability for pollution dispersion modeling	National (Meteorological Department)
	Health impact data	Respiratory disease cases linked to air pollution, sandstorm-related health outcomes	National (Ministry of Health)
	Ozone layer data	Ozone-depleting substances consumption, phase-out plan implementation data	National (EPA), International (Montreal Protocol, UNEP, UNIDO)
7. Biodiversity	Species inventory data	Number of species by classification (plankton, annelids, cnidarians, mollusks, arthropods, chordates, plants, etc.)	National (KISR, Kuwait University, IUCN)
	Habitat data	Distribution and condition of terrestrial, coastal, and marine	National (EPA, PAAF, KISR)

		habitats, nature reserves (area, number)	
	Protected areas data	Area of natural reserves (km ²), management effectiveness, conservation outcomes	National (EPA)
	Threat data	Urbanization impacts, grazing pressure, overfishing, pollution impacts, invasive species	National (EPA, KISR)
	Aichi Biodiversity Targets data	Progress indicators for each of the 20 Aichi Targets	National (EPA, CBD reporting)
	Fisheries data	Fish and shrimp landings, stock assessments, fishing effort, breeding cycles	National (PAAF, KISR)
8. Coastal and Marine Environment	Marine water quality data	Pollution levels (industrial discharge, oil spills, desalination plant brine), eutrophication indicators, nutrients	National (EPA, KISR)
	Coastal zone data	Coastal land use, urbanization, dredging and reclamation activities, artificial islands	National (EPA, Kuwait Municipality)
	Oil pollution data	Oil spill incidents, volumes, cleanup operations, impacts on marine life	National (Kuwait Oil Company, EPA)
	Invasive species data	Ballast water discharge, invasive species presence, spread rates, ecological impacts	National (KISR, EPA)
	Marine biodiversity data	Coral reef health, mangrove cover (e.g., 0.58 km ² in 2017), fish mortality incidents	National (KISR, EPA)
	Sea level rise monitoring data	Sea level measurements from monitoring stations (e.g., Salmiya station operational 2023)	National (KISR)
	Fisheries management data	Fish and shrimp stock assessments, sustainable catch limits, aquaculture projects	National (KISR, PAAF)
9. Environmental Policies	Legislative and regulatory data	Environmental Protection Law No. 42 of 2014 and amendments, executive regulations, compliance mechanisms	National (EPA, legislative bodies)

	Institutional framework data	Roles and responsibilities of EPA, other government entities, environmental departments within institutions	National (EPA, Council of Ministers)
	International commitments data	UNFCCC submissions (national communications, biennial update reports, NDCs), Paris Agreement, CBD, UNCCD, Ramsar, etc.	National (EPA), International (UNFCCC, CBD, etc.)
	Climate action data	National Adaptation Plan (2019-2030), low-carbon development strategy, emission reduction targets (7.4% by 2035), carbon neutrality target (2060)	National (EPA, Supreme Council for the Environment)
	Urban environmental data	Air quality strategy (2040), waste management strategy (2040), energy efficiency plans, green spaces	National (EPA, MEW, Kuwait Municipality)
	Economic instruments data	Pollution taxes, licensing systems, financial incentives, feed-in tariffs, subsidies, customs exemptions	National (Ministry of Finance, EPA)
10. Future Scenarios	Scenario planning data	Driving forces of environmental change, uncertainty assessments, stakeholder perspectives	National (stakeholder surveys, 26 participating organizations)
	Policy impact data	Potential impacts of carbon taxes, subsidy reforms, renewable energy incentives, circular economy transitions	National (EPA, academic studies, scenario modeling)
	Economic diversification data	Investment trends, private sector participation, green jobs, renewable energy industries, waste recycling industries	National (CSB, Ministry of Commerce and Industry, PAI)

6.1.2 Categories of Data Source in Kuwait SOE Report 2024

Table 10 Categories of Data Source in Kuwait SOE Report 2024

Source Category	Examples from Report
National Statistical Agencies	Central Statistical Bureau (CSB), Environment Public Authority (EPA)
Line Ministries and Government Entities	Ministry of Electricity, Water and Renewable Energy (MEW), Ministry of Health, Ministry of Public Works, Ministry of Interior, Kuwait Municipality, Public Authority for Industry (PAI), Public Authority for Agriculture Affairs and Fish Resources (PAAF), General Department for Information Systems
Research and Academic Institutions	Kuwait Institute for Scientific Research (KISR), Kuwait University, Arabian Gulf University
State-Owned Enterprises	Kuwait Oil Company (KOC), Kuwait Petroleum Corporation (KPC)
International Organizations	UNEP, UNDP, WHO, UNFCCC, IEA, IUCN, UNIDO, FAO
Peer-Reviewed Literature	Academic journals (<i>Frontiers in Environmental Science, Sustainability, Journal of Natural Gas Science and Engineering, etc.</i>)
Primary Data Collection	EPA air quality monitoring network, EPA waste management database project, KISR marine monitoring stations, stakeholder surveys for scenario planning

6.1.3 Identified Data Gaps in Kuwait SOE Report 2024

Table 11 Data Gaps in Kuwait SOE 2024

Data Gap Description	Chapter	Context / Source
No data on key air pollutants (NO _x , SO ₂) from oil, gas, and petrochemical sector	Chapter 6: The Atmosphere	The report states: "The oil, gas and petrochemical sector calculates carbon emissions only, and no data is available on emissions of other key gaseous pollutants such as nitrogen oxides or sulfur dioxide." This represents a critical gap for comprehensive air quality management.
Gaps in air quality monitoring systems, information systems,	Chapter 6: The Atmosphere	The EPA is preparing a national air quality strategy to "fill many of the current gaps in air quality management, such as completing monitoring systems,

and forecasting programmes		information systems and forecasting programmes, and conducting environmental load reporting."
Lack of historical comprehensive waste management database	Chapter 3: Waste Management	The report notes that a project titled "Survey and establishment of a comprehensive waste management database in Kuwait" was implemented to collect data on sources, types, properties, and quantities of waste. This implies that such a database did not previously exist in a comprehensive form.
No system for waste segregation at source; recycling conducted randomly	Chapter 3: Waste Management	The report states: "Although 24 recycling plants are approved by the EPA, no system is in place to segregate solid waste from its source. Instead, sorting and recycling are conducted randomly." This indicates a lack of systematic data on recycling streams and outcomes.
Lack of data on landfill gas, groundwater, and odor monitoring (historically)	Chapter 3: Waste Management	The report mentions that "In all landfills, a comprehensive monitoring programme for landfill gases, groundwater, solid waste and odours has been implemented, and an environmental risk assessment has been conducted." This suggests such monitoring was previously lacking.
Missing data on emissions of regulated air pollutants from power and water plants	Chapter 6: The Atmosphere	While the report cites increased fossil fuel consumption and resulting emissions, it notes that data on specific pollutants (SO ₂ , NO _x) from power plants is not consistently reported or integrated, contributing to gaps in the national emissions inventory.
Incomplete marine and coastal environment monitoring infrastructure	Chapter 9: Environmental Policies	The report explicitly states: "It is necessary to complete the construction of infrastructure for the monitoring and control of the marine and coastal environment and to finalize databases for the proper management of marine resources in support of the blue economy."
Lack of unified biodiversity database or time-series trend data	Chapter 7: Biodiversity	While the report provides a species count (Table 1, page 22), it lacks time-series data on population trends, species decline rates, or ecosystem health indicators. The Aichi Biodiversity Targets progress table (Table 2, page 23) indicates incomplete data for several targets, particularly related to monitoring and reporting.
Gaps in climate action policies and related data for monitoring	Chapter 9: Environmental Policies	The report notes: "It is therefore necessary to fill the current gaps in climate action policies to fulfil the obligations of Kuwait as a State party to the UNFCCC and the Paris Agreement, and achieve SDG 13." This

		includes gaps in data needed to track mitigation and adaptation progress.
Lack of data on agricultural water use efficiency and soil salinity impacts	Chapter 5: Water Resources	The report cites inefficient use of brackish water in Al-Wafra agricultural area between 1995 and 2006 but lacks recent quantitative data on irrigation efficiency, soil salinity trends, or the relationship between water use and agricultural productivity.
No centralized or integrated environmental data management system	Chapter 9: Environmental Policies	The report notes that while the Environmental Protection Law No. 42 of 2014 provides for environmental data management systems, such systems are not yet fully operational or integrated across government entities.
Lack of data on transboundary dust storm sources and composition	Chapter 4: Land Resources	While the report identifies sand and dust storms as a major issue, it notes limited data on the contribution of cross-border sources (e.g., from Iraq) and the chemical composition of dust particles, which affects public health assessments.
Incomplete data on wastewater reuse and discharge volumes	Chapter 5: Water Resources	The report states: "Treated wastewater is an important alternative to limited fresh water, yet reused wastewater accounted for only 36 per cent of the volume of treated wastewater in 2018." It also notes that more than 120 million cubic meters of liquid waste is discharged into the sea annually, but detailed characterization of this discharge is lacking.
Lack of data on the health impacts of air pollution and sandstorms	Chapter 6: The Atmosphere	While the Ministry of Health conducted studies linking respiratory diseases to dust waves, the report indicates that systematic, longitudinal health impact data linked to air quality monitoring is still under development.
No data on the economic costs of environmental degradation	Multiple chapters	Throughout the report, economic impacts are mentioned qualitatively (e.g., "economic loss" from sandstorms, page 14; "economic losses associated with cleanup operations" from oil spills, page 22), but quantified economic data on the costs of environmental degradation is absent.
Lack of integrated data across water-energy-food nexus	Chapter 9: Environmental Policies	The report recommends incorporating the interconnectedness of water, energy, and food sectors in policy development, implying that integrated datasets linking these sectors are currently insufficient or non-existent.

Limited data on invasive species and ballast water impacts	Chapter 8: Coastal and Marine Environment	The report identifies invasive alien species from ballast water as a serious threat but provides no quantitative data on the number of invasive species, their spread rates, or their ecological and economic impacts.
Gaps in environmental impact assessment (EIA) data and compliance monitoring	Chapter 9: Environmental Policies	The report notes the existence of an environmental and social impact assessment system but does not provide data on the number of EIAs conducted, compliance rates, or the quality of follow-up monitoring.
Lack of data on private sector and non-governmental environmental contributions	Chapter 7: Biodiversity	The report mentions that "17 governmental organizations, as well as volunteer groups and private sector companies, are involved in achieving the goals of the National Biodiversity Strategy," but no data is presented on the scope, scale, or outcomes of these contributions.

6.1.4 Analysis of SOE 2024 Data Sources

The Kuwait SOE Report Data highlights a strategic focus on national-level sources, supported by National Reports & national Strategies which combined contributed to around 39.8% of the report data sources. There is significant reliance on international sources particularly Conventions and Protocols in addition to United Nations and other International Organizations. Regional organizations, peer-reviewed journals, and academic institutions provide specific data to support the broader regional context and strategic environmental categories.

Key Observations

Observation	explanation
High percentage of National Sources	National Government related Institutions and National Reports & Strategies together account for 39.8% of all source types. This highlights that the report depends on profoundly on domestic data and national policy documents.
Strong International Framework	International Conventions & Protocols, UN & International Organizations, and International Reports & Frameworks collectively represent 39.8% of sources, reflecting Kuwait's engagement with global environmental governance.
Moderate Research Input	National Research & Academic Institutions and International Peer-Reviewed Journals represent only 12% of sources, suggesting that academic literature plays a supporting rather than primary role in the report's data foundation.
Limited Regional Sources	Regional Organizations & Conventions account for only 6% of sources, indicating that GCC-level or regional data sharing is underrepresented compared to national and international sources.

Underutilized International Research	International Academic & Research Institutions represent the smallest category (2.4%), despite the report mentioning a collaboration with Harvard School of Public Health for air quality work.
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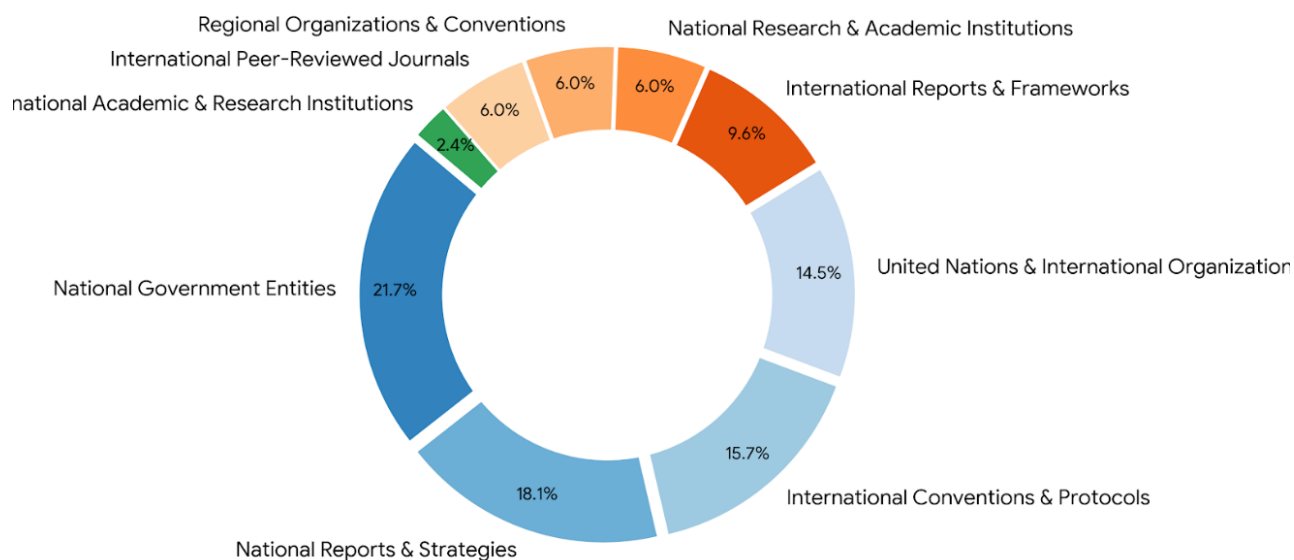
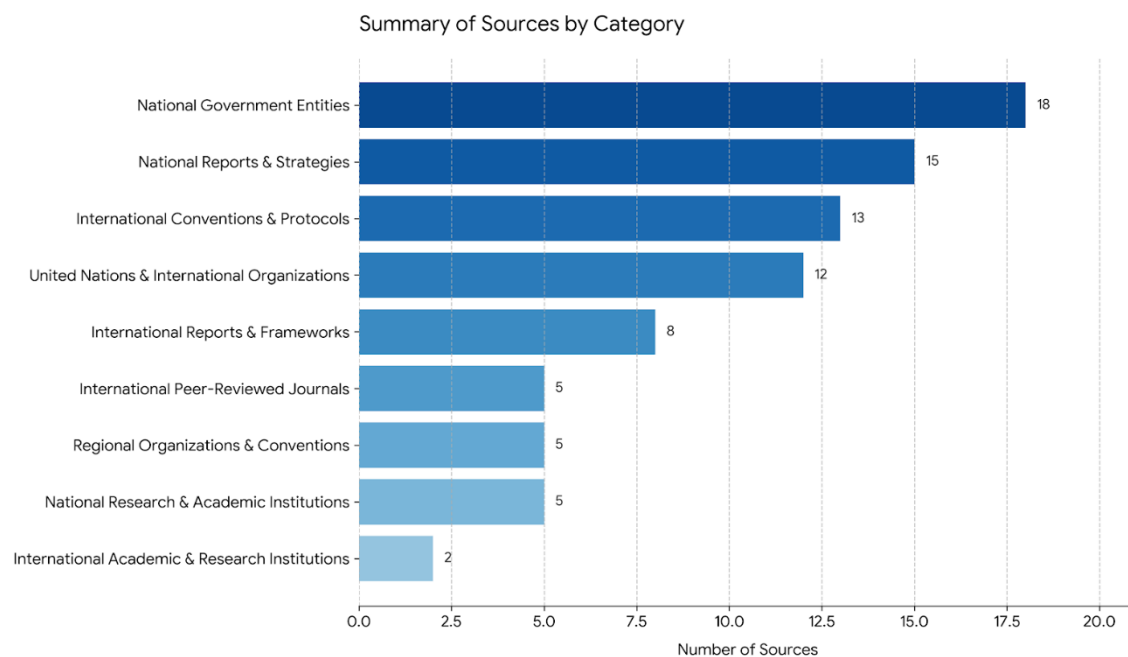


Figure 7 Distribution of Data Sources by Category for Kuwait SOE Report

6.1.5 Overview of Multilateral Agreements in the Report

Kuwait SOE report 2024 has identified more than 20 multilateral environmental agreements (MEAs) in multiple chapters. These references serve several objectives, particularly to:

- Demonstrate Kuwait's compliance with international obligations
- Provide the legal and policy framework for national environmental actions
- Identify gaps in implementation, assessment and reporting
- Align Kuwait national strategies with international conventions and frameworks (e.g., SDGs, Aichi Targets)

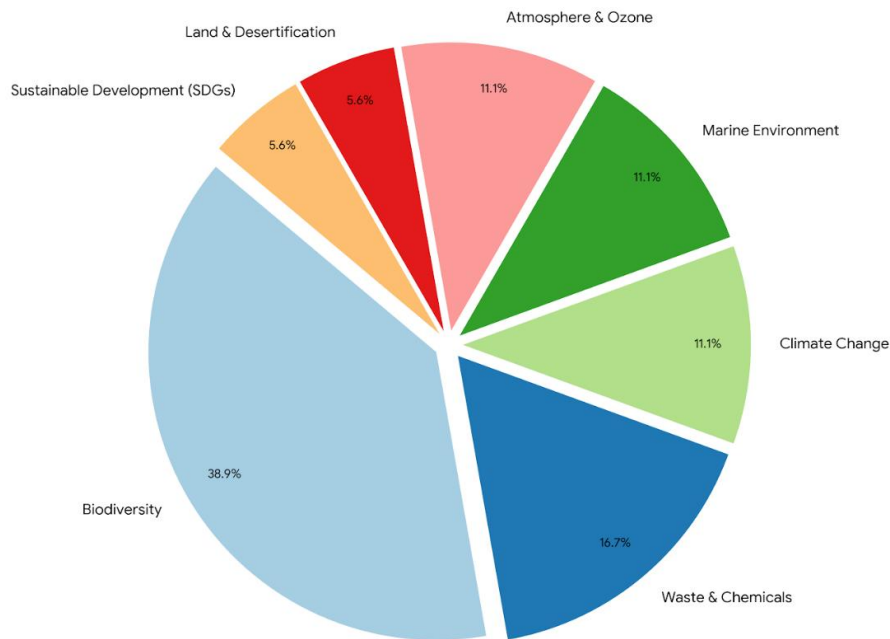


Figure 8 SOE Multilateral Agreements by Theme

7 Assessment of Environmental Data Gaps

Kuwait has a diverse wealth of data and specialized environmental institutions with vast expertise. However, there are wide gaps and lacks in certain environmental domains including thematic scope, coordination between institutions, data management, data accuracy and consistency, data sharing, and citizen awareness. Therefore, the National Environmental Data Management Plan (NEDMP) aims to minimize these lacks through the identification of these gaps and by establishing regulatory frameworks, harmonizing environmental data and its aspects. NEDMP aims at filling these gaps in accordance with global standards through enabling information communication between systems, and establishing community awareness in addition to boosting expertise in institutions dealing with environment data. Accordingly, the framework provides the enabling environment to turn the scattering of data in Kuwait into a national integrated information system for the environment in accordance with Kuwait Vision 2035.

7.1 Thematic Data Gaps

Table 12 Thematic Data Gaps

Thematic Area	Identified Gap	Comments
Air Quality	Limited coverage beyond urban/industrial centers; lack of integrated national emissions inventory	eMISK covers air quality primarily from EPA monitoring stations; no comprehensive national emissions inventory documented
Climate Change	Insufficient historical and projected climate data; limited greenhouse gas inventory granularity	References to NAPs and NDCs exist but detailed climate modeling data gaps not addressed
Biodiversity	Incomplete species inventories; lack of systematic wildlife monitoring data	Planned VGI wildlife sightings initiative indicates current gap; species data fragmented across KISR, PAAAFR, EPA
Groundwater	Limited real-time monitoring; fragmented data between MEW and KISR	Water resources data noted but not systematically integrated
Soil and Land Degradation	No centralized soil quality database; desertification indicators incomplete	Soil data mentioned but not part of eMISK core domains; Kuwait Municipality holds some land use data
Marine Benthic Habitats	Incomplete mapping of seabed habitats and coral reef distributions	KISR conducts research but no continuous monitoring system documented
Waste Management	Hazardous waste tracking data incomplete; limited landfill monitoring data	Waste management data exists but not fully integrated across all waste streams
Environmental Health	Weak linkage between environmental quality data and public health outcomes	Ministry of Health data exists separately from EPA monitoring; no integrated environmental health surveillance system

7.2 Institutional and Coordination Gaps

Table 13 Institutional and Coordination Gaps

Gap Category	Description	Evidence
Fragmented Marine Data	Both EPA and KISR maintain marine environmental data with potential duplication and inconsistent standards	EIS paper (KISR, 2002) focused on marine; eMISK includes marine quality; no single authoritative marine data source identified

Missing EPA Integration	EPA not explicitly referenced in Kuwait Data Map (2020), suggesting environmental data not fully integrated into national data infrastructure	Kuwait Data Map lists government entities but does not mention EPA or specialized environmental datasets
Duplication of Effort	Overlap in industrial effluent monitoring across EPA, KISR, and SAA-EPC	Multiple institutions collecting similar data without clear coordination mechanisms
Limited Data Sharing Agreements	No formalized data sharing protocols documented between EPA and sectoral ministries	NEDMP objectives explicitly target interoperability, indicating current gaps
Weak Academic-Government Linkages	Research data from Kuwait University, KISR not systematically integrated into EPA's operational systems	Academic institutions listed as separate sources; no integration framework mentioned

7.3 Data Lifecycle Gaps

Table 14 Data Lifecycle Gaps

Lifecycle Stage	Gap	Alignment with NEDMP Objectives
Plan	Absence of comprehensive data management plans across institutions	SO1: Institutionalize Governance
Acquire	Inconsistent data collection methodologies; no unified national sampling protocols	SO2: Quality & Standards
Process	Limited metadata standardization; QA/QC frameworks not uniformly applied	SO2: Quality & Standards
Analyze	Insufficient analytical capacity for integrated environmental assessments	SO3: Interoperability
Preserve	No centralized long-term environmental data archive; risk of data loss	SO5: Capacity Building
Publish/Share	Limited open data access; Beatona provides public access but not all data available	SO4: Accessibility & Transparency

7.4 Geographic and Temporal Coverage Gaps

Table 15 Geographic and Temporal Coverage Gaps

Gap Type	Description
Geographic Coverage	Monitoring concentrated in urban and industrial areas (Kuwait City, Shuaiba, Ahmadi); limited coverage of remote desert areas, offshore marine zones, and islands
Temporal Coverage	Time series data gaps for long-term trend analysis; pre-1990 environmental baseline data largely unavailable; post-Gulf War recovery data incomplete
Real-Time Monitoring	Limited real-time data availability beyond selected air quality stations; water quality and marine data often rely on periodic sampling rather than continuous monitoring

7.5 Data Standardization and Quality Gaps

Table 16 Data Standardization and Quality Gaps

Gap	Description
Metadata Standards	No unified metadata framework across institutions; data discovery and interoperability limited
QA/QC Protocols	Variable quality assurance practices across data producers; no centralized quality control mechanism
Data Formats	Inconsistent data formats hinder integration; eMISK uses GIS layers but other institutions use disparate systems
Unit Consistency	Lack of standardized units and measurement protocols across monitoring programs

7.6 Accessibility and Transparency Gaps

Table 17 Accessibility and Transparency Gaps

Gap	Description
Restricted Data Access	Not all environmental data available through Beatona; some datasets remain internal to institutions
Language Barriers	Limited Arabic-language accessibility for technical environmental data

User Awareness	Low public awareness of available environmental data and how to access it
Data Usability	Raw data often requires technical expertise to interpret; limited user-friendly visualization tools beyond Beatona

7.7 Capacity and Sustainability Gaps

Table 18 Capacity and Sustainability Gaps

Gap	Description
Human Resources	Insufficient trained personnel for data management, analysis, and interoperability implementation
Technological Infrastructure	Aging monitoring equipment; need for modernization of data collection and management systems
Sustainable Funding	Reliance on project-based funding; no dedicated long-term funding mechanism for environmental data systems
Training Gaps	Limited training programs in geo-environmental analysis and data science across institutions

Table 19 Categorisation of Environmental Data Gaps

Category	Priority Gaps
Thematic	Biodiversity, groundwater, soil degradation, climate change indicators, environmental health linkages
Institutional	EPA integration into national data infrastructure; formalized data sharing agreements; duplication resolution
Data Lifecycle	Metadata standards; QA/QC frameworks; centralized archiving; open data publishing
Geographic/Temporal	Remote area coverage; long-term time series; real-time monitoring expansion
Standardization	Unified metadata; consistent QA/QC; standardized formats and units
Accessibility	Open data policy; user-friendly tools; Arabic language accessibility
Capacity	Trained personnel; sustainable funding; modern infrastructure

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